

Vinh Ly

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Summary

Software Engineer with **18 years** designing and owning distributed systems at scale. Built data pipelines processing billions of events daily and migrated services handling hundreds of millions of users at Google, YouTube, and Whatnot. Led teams of 5-8 engineers through end-to-end delivery of high-traffic infrastructure. Expert in distributed systems, backend architecture, and kernel-level performance optimization. Seeking to apply deep ownership mindset to complex backend challenges at a high-growth startup. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS

Distributed Systems: Spanner, Bigtable, BigQuery, MapReduce, Kafka, Pub/Sub, Cassandra

Backend Engineering: Go, C++, Python, Java, SQL, REST APIs, gRPC, Protocol Buffers

Infrastructure & Reliability: Kubernetes, Docker, GCP, on-call ownership, SLO/SLA design, postmortems

Data Engineering: ETL pipelines, data warehousing, streaming ingestion, real-time analytics

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Own core backend infrastructure serving hundreds of millions of daily active users with direct P0 on-call ownership.
- Redesigned data ingestion pipeline reducing end-to-end latency by 40% while handling 10x traffic growth.
- Led migration of analytics services from legacy MapReduce jobs to streaming architecture on Kubernetes.
- Mentored 4 engineers on distributed systems design patterns and performance debugging techniques.
- Established SLO framework for critical data paths reducing customer-facing incidents by 35%.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Optimized database query patterns on Spanner cutting p99 latency from 800ms to 120ms for listing searches.
- Designed streaming pipeline for real-time inventory updates serving 50K+ concurrent sellers during peak events.
- Architected multi-region data synchronization layer for marketplace transaction processing with strong consistency guarantees.
- Led incident response and blameless postmortems for critical path failures affecting platform availability.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Migrated 200+ internal services from legacy infrastructure to Kubernetes resulting in 60% infrastructure cost reduction.
- Managed team of 8 engineers owning data infrastructure for core Google product telemetry ingestion.
- Built real-time anomaly detection system on BigQuery processing 500M events daily with sub-minute alerting.
- Led quarterly architecture reviews and mentored senior engineers on systems design and career growth.

- Designed distributed locking mechanism for cross-region data consistency in Spanner at 100K TPS.

Software Engineer | Google | 02/2011 - 02/2015

- Optimized Bigtable access patterns cutting storage costs by \$2M annually while improving read throughput.
- Developed backend services for Google Maps infrastructure processing location data from 1B+ devices.
- Implemented data validation framework catching pipeline errors before production deployment.
- Authored technical documentation and onboarding guides for team knowledge sharing.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed backend systems for Azure table storage handling millions of customer requests per second.
- Conducted performance testing and capacity planning for cloud storage infrastructure at scale.
- Implemented monitoring dashboards providing visibility into system health and performance metrics.
- Collaborated with kernel-level teams on storage subsystem optimizations in Windows Server.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Built internal tooling for automated testing of distributed storage systems.
- Contributed to design documentation for system reliability improvements.

Education

B.S. Honors

Virginia Tech | 2004 - 2007